The Ford-Damped Operator Family: The Boundary Re s = 1/2, a Totient Moment, and an Eisenstein-Coefficient Comparison

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November 2022

Abstract

We introduce the Ford-damped family of Farey-graph operators $L(s)$, $s \in \mathbb{C}$, with edge weights $(q_u q_v)^{-2s} = e^{-s\cdot(\text{sum of horoball depths})}$; the canonical operator of the companion saturation-law paper is $L(1)$. We prove three anchor theorems and report one measured phenomenon; everything else is an explicitly open program, and no part of RH is claimed.

Theorem A (existence dichotomy and zeta norm) [proved]. The absolute-weight form associated with $L(s)$ is bounded on the finitely supported core for $\mathrm{Re}\ s > 1/2$; for real $s \leq 1/2$, the positive form is $+\infty$ on every nonzero core vector. Thus $\mathrm{Re}\ s = 1/2$ is the boundary of this bounded-operator construction, not a theorem about zeta zeros. For $\mathrm{Re}\ s > 1/2$, $\|L(s)\| \leq 4\zeta(2\ \mathrm{Re}\ s)$; the bound diverges with ζ's pole as $\mathrm{Re}\ s \downarrow 1/2$, and at $s = 1$ it reproduces the canonical bound $2\pi^2/3$ of the companion operator paper.

Theorem B (Hurwitz diagonal) [proved]. For $q \geq 2$, the diagonal is $W_s(p/q) = q^{-4s}[\zeta(2s, a/q) + \zeta(2s, (q-a)/q)]$ with $ap \equiv 1 \pmod q$; for $q=1$ it is $2(\zeta(2s)-1)$. Hence every matrix entry of the full-weight truncation $L_N(s)$ continues meromorphically to $\mathbb{C}$ with a single simple pole at $s = 1/2$, of residue q^{-2} on the diagonal - the boundary obstruction is absolute summability on the stated Hilbert-space core, not entrywise algebra.

Theorem C (rescaled boundary limit) [proved]. For each fixed N, $\lim_{s\downarrow 1/2}(2s-1)L_N(s) = 2\cdot\mathrm{diag}(q^{-2})$ in norm (limit along real s): a multiplication operator whose eigenspaces are denominator shells, with eigenvalue $2/q^2$ of multiplicity $\phi(q)$ (under the adopted periodic mod-1 vertex convention; the source computation's two-endpoint convention has one additional $q = 1$ vertex). Letting $N \to \infty$ after $s \downarrow 1/2$, the positive-power moment trace of $A_N := T_N/2$ converges for $\mathrm{Re}\ w > 1$ to $\Sigma\ \phi(q)q^{-2w} = \zeta(2w-1)/\zeta(2w)$, locally uniformly in w. This ratio is the finite-prime factor in the modular Eisenstein coefficient evaluated at the distinct variable w; it is not the full coefficient and the family boundary $s = 1/2$ is not being substituted into that coefficient. Equivalently, the limit is the spectral zeta of the unbounded shell operator H_∞ defined in Corollary 6.2. The inverse-power trace of A_N instead has terms q^{2w}. The Dirichlet-series identity is classical; the contribution claimed here is only the stated operator-family route to the finite sums and their ordered limit.

[Measured; supplementary provenance only]. At $N=60$, a pinned package-local experiment on the 1102-vertex periodic graph truncation finds that the normalized bottom-projector IPR decreases from 0.5000 at $s=1$ to 0.1116 at $s=0.51$, while λ_2 increases from 6.187×10^{-7} to 7.761×10^{-4}. The separately preserved 1103-vertex two-endpoint source experiment reports single-eigenvector IPR $0.5000 \to 0.0593$ and $\lambda_2=6.187\times10^{-7} \to 7.620\times10^{-4}$ at the same sampled s values. These are distinct finite graph truncations and distinct localization diagnostics; neither is the periodic full-weight compression $L_N(s)$. Theorem C supplies a candidate shell mechanism, but no transfer, finite-size-scaling, or $N\to\infty$ claim is made.

The family is compared at $s = 1$ with a separate companion magnetic construction. At the family boundary, its ordered shell-moment limit is compared with the finite-prime factor of an Eisenstein coefficient in the independent variable w. Section 7 formulates bounded operator questions on real $s \in (1/2, 1]$, with sub-questions Q1-Q5 and failure criteria F1-F3. No result in this paper turns those questions into an RH criterion or connects an answer to the truth of RH. The anchor theorems are priority results C-26, C-27, C-28 of the Branch C ledger (packets BC-015, BC-016, BC-018).

Scope and evidence boundary

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Prior public version: Zenodo record 21231887, DOI 10.5281/zenodo.21231887
 Authorial research provenance: The underlying SIT and Branch C research program was developed by Micah Blumberg across 2011–2026, with its core conceptual foundations established before the public release of ChatGPT in November 2022. AI did not originate the theory, hypotheses, research program, or source genealogy. Later AI tools supplied bounded assistance with mathematical formulation, verification code, source checking, grammar, and editorial review under the author's direction; those outputs are not treated as independent scientific evidence.
Claim boundary: No claim or implication of RH. The proved content is an existence/no-go dichotomy (Theorem A / 4.4), a norm bound with tracked constants (Theorem A(a)), exact identities (Theorem B / 5.1), and a rescaled boundary limit (Theorem C / 6.1); the operator questions of §7 are open mathematics with pre-registered failure criteria. Neither RH nor full physical SIT is claimed proved anywhere in this paper. **Evidence-class convention (used throughout).** Every claim in this paper carries one of three labels: **[proved]** - a theorem proved in this paper or imported with citation; **[measured]** - a finite numerical observation with named script, packet, and parameters, with no asymptotic claim; **[conjectural]** - an open question or program item, asserted nowhere as fact. **Operator-power convention (used throughout).** Put $A_N := T_N/2 = \mathrm{diag}(q_v^{-2})$ under the periodic mod-1 vertex convention fixed in Definition 3.1. **[1]** For every fixed N and $w \in \mathbb{C}$, powers use the real logarithm of the strictly positive eigenvalues. Thus $\mathrm{Tr}(A_N^w) = \Sigma_{q\le N} \phi(q)q^{-2w}$, whereas $\mathrm{Tr}(A_N^{-w}) = \Sigma_{q\le N} \phi(q)q^{2w}$. **[2]** We call the first expression the **positive-power moment trace**: "positive" names the sign of the operator exponent, not the sign of w or of the trace. Equivalently, for finite N only, define the inverse shell matrix $H_N := A_N^{-1} = \mathrm{diag}(q_v^2)$; then the same finite sum is its conventional spectral zeta $\zeta_{H_N}(w) := \mathrm{Tr}(H_N^{-w})$. **[3]** This inverse is a deliberate reparameterization, not a claim that A_N itself has that spectral zeta and not an additional connection to scattering theory. We never identify $\mathrm{Tr}(A_N^{-w})$ with the decaying totient polynomial.

1. Introduction

1.1 The problem and operation before the name

Consider a graph whose vertices are rational points and whose edge weights decrease as the denominators grow. Three questions arise before any specialized name is needed. First, for which damping strengths does the infinite weighted Laplacian define a bounded operator? Second, what exactly survives when the damping approaches its boundary value? Third, can the surviving denominator shells be related to a classical arithmetic function without confusing a finite matrix identity, an ordered limit, and a full scattering theory?

Established analysis supplies the parts: Schur-type operator bounds, Ford horoball geometry, Hurwitz-zeta continuation, the totient Dirichlet series, and the finite-prime factor of the Eisenstein coefficient. The operation studied here is to turn hyperbolic penetration depth into edge damping, prove the boundedness boundary, rescale the full-weight finite compression at that boundary, read the resulting denominator shells, and only then take their positive-power moment limit.

The name **Ford-damped operator family** abbreviates that operation. It is not a new name for an established transfer operator or Eisenstein scattering operator. The construction fails to support its intended interpretation if the infinite operator is unbounded in the claimed region, if the rescaled finite compression lacks the stated shell limit, if the moment trace uses the wrong operator power, or if the order of limits is reversed without a new proof. These failure conditions determine the claim ceiling throughout.

1.2 Motivation

Two zeta-related constructions motivate two comparisons. At `s = 1`, companion Papers 1/3 study a separate magnetic deficit motivated by Mertens-type quantities; this paper neither imports its claimed equivalence nor uses it as a theorem. Independently, the modular Eisenstein coefficient is Φ_Eis(w)\=\Λ(2w-1)/Λ(2w) in its own variable `w` [classical]. The operator family reaches the boundary `s = 1/2`, where an ordered limit of its shell moments gives only the finite-prime zeta ratio inside `Φ_Eis(w)`. The two variables and the two statements are kept distinct.

This paper exhibits a one-parameter operator family connecting them. The weights are not chosen to make the connection work: they are forced by hyperbolic geometry. Each Farey fraction `p/q` carries a Ford horoball of depth `2 log q` below the standard cusp horoball, and the edge weight `(q_u q_v)^{-2s}` is exactly `e^{-s·(sum of the two horoball depths)}` (Proposition 3.3) [proved]. The family `L(s)` is therefore the Farey Laplacian damped by hyperbolic penetration depth, with `s` the damping rate.

Three structural facts, proved here, distinguish the family from an arbitrary interpolant:

1. its **bounded-operator boundary is the line** `Re s = 1/2` - the absolute-weight form is finite on the core for `Re s > 1/2`, while the positive form diverges for real `s ≤ 1/2` (Theorem A) [proved];
2. its **entrywise meromorphic continuation has a single simple pole at** `s = 1/2` with residue `q^{-2}` on the diagonal, so the boundary obstruction is absolute summability on the stated core, not entrywise algebra (Theorem B) [proved];
3. its **rescaled degeneration at that boundary is exactly solvable**, with totient-shell eigenspaces and positive-power moment limit `ζ(2w-1)/ζ(2w)` - equivalently, the spectral zeta of a separately defined unbounded inverse-shell operator, and only the finite-prime factor of `Φ_Eis(w)` (Theorem C and Corollary 6.2) [proved].

Section 7 asks only whether the two endpoint descriptions admit useful uniform operator estimates [conjectural]. No theorem, criterion, or implication concerning RH follows from the endpoint comparison or from the open questions posed here.

The program map is deliberately one-way. At `s=1` lies the canonical phase-free operator, alongside the separate magnetic construction of companion Papers 1/3. At the boundary lies the fixed-`N` rescaled shell operator and, only after that limit, its positive-power moment in the independent variable `w`. Q1-Q5 ask whether controlled estimates connect these descriptions; F1-F3 are route-specific exits. Every connection in this map is open rather than a proved implication.

1.3 Main results

The results are stated informally in the abstract; the full statements with hypotheses are Theorem 4.4 (= Theorem A), Theorem 5.1 (= Theorem B), Theorem 6.1 with Corollary 6.2 (= Theorem C), and Supplementary Measurement 6.5 (the two-convention finite experiment, which is evidence, not a theorem). The abstract states each at full strength; nothing stronger is claimed in the body.

1.4 Roadmap

§2 places the construction against prior work (transfer operators, the Farey spin chain, Eisenstein coefficients, Farey statistics) and states the logged novelty claim. §3 fixes definitions: the Farey graph, Ford weights, the family `L(s)` and its core, and the two truncations `L_N(s)` (graph truncation, entire entries) and `L̃_N(s)` (full-weight truncation, carries the pole) - a distinction that matters everywhere in §§5-6. §4 proves the existence dichotomy and the `4ζ(2σ)` norm bound with all constants tracked. §5 proves the Hurwitz-diagonal identity, the entrywise continuation, and norm-holomorphy of the family. §6 proves the rescaled boundary limit, derives the positive-power moment identity with the order of limits explicit, and reports the measured delocalization data. §7 formulates the operator problem as five precise questions Q1-Q5 with failure criteria F1-F3, and §7.6 states the scope and limitations of the whole paper. Appendix A fixes conventions and lists the machine-verification inventory.

2. Relation to prior work

Positioning per the series prior-art review (production-canon/papers/branch-c-series/wiki/PRIOR-ART-REVIEW-20260612.md); the live review log is cluster page S07.

Transfer operators (Mayer; Lewis-Zagier; Farey-map school). Mayer expresses the Selberg zeta of $\mathrm{PSL}(2,\mathbb{Z})$ as a Fredholm determinant of the Gauss-map transfer operator; Lewis-Zagier put Maass forms in bijection with solutions of the three-term functional equation; the Farey-map strand (Prellberg; Bonanno-Isola, organizing transfer-operator iterates by the Stern-Brocot tree) is the closest established school connecting Farey structure to modular spectra. Our family is a graph Laplacian on $\ell^2(\mathfrak{F})$ (self-adjoint for $s \in \mathbb{R}$), not a composition-type transfer operator on a function space. [Measured] evidence of difference: the bottom eigenvector of the source two-endpoint matrix `L_N^{2e}(1)` is uncorrelated with the Gauss-Kuzmin density ($r = -0.06$ at $N = 60$; packet BC-013, script bc013_transfer_operator_probe.py) - a single-size numerical exclusion, not a proof of non-conjugacy. Whether $L(s)$ is conjugate or intertwined with a known transfer operator is open [conjectural], together with a distributional framework for $\sigma \leq 1/2$ (BC-GAP-3b).

The Farey spin chain (Knauf; Kleban-Özlük; Fiala-Kleban). The number-theoretic spin chain has a phase transition at $\beta = 2$, in the same `(q_uq_v)`-exponent family as our weights. Whether Theorem A's dichotomy is the Knauf transition in graph-Laplacian clothing is open [conjectural]; the objects differ (free energy of a partition function vs. form-boundedness over a core), but the coincidence of critical exponents is flagged here deliberately. An independent tradition meets the same critical point as our C-27 boundary.

Eisenstein coefficients and RH reformulations. The modular coefficient Φ_Eis(w)\=\Λ(2w-1)/Λ(2w) and the spectral decomposition of the modular surface are classical (Selberg; Lax-Phillips; Iwaniec; Hejhal). Lax-Phillips theory contains its own RH reformulations; Theorem C is not one of them. Corollary 6.2 produces only the finite-prime factor $\zeta(2w-1)/\zeta(2w)$ after a separately ordered operator limit, and that Dirichlet-series identity is classical (DLMF 27.4.6; Hardy & Wright). Closest precedent for critical-strip zeta from finite graph spectra: Friedli-Karlsson (torus graphs; Eisenstein connection noted there); our contribution is the Farey-Laplacian route, not the destination identity. Herichi-Lapidus exhibit a different operator family whose invertibility transition also sits at the critical line; mechanism and carrier differ.

Farey statistics and graph spectra. Marklof's *Horospheres and Farey fractions* (arXiv:1003.4613) and Athreya-Cheung's Poincaré-section analysis (doi:10.1093/imrn/rnt003) are primary sources for the horocycle-flow treatment of Farey statistics relevant to possible $N \to \infty$ work on Q1-Q4. Franel-Landau (RH $\Longleftrightarrow$ Farey discrepancy; modern treatment Kanemitsu-Yoshimoto) underlies the graph's RH pedigree and is used in Paper 4, but no Franel-Landau statement is imported into a proof here. Unweighted finite Farey-graph Laplacian spectra appear in network science (Zhang-Comellas) with no arithmetic weights; within the literature reviewed (logged in the prior-art review), the weighting `(q_uq_v)^{-2s}` does not appear there.

Hilbert-Pólya boundary. We make no Hilbert-Pólya claim: the family is a carrier of equivalences and limits, not a proposed zero-spectrum operator. Nothing in this paper, and nothing in the literature reviewed, proves RH.

Novelty summary (logged review; scope-limited). Within the literature covered by the logged review, no publication was found that builds a weighted Laplacian on the Farey graph with Ford / horoball damping, proves an existence dichotomy in the weight exponent, or obtains the finite-prime Eisenstein zeta ratio as an ordered moment limit of such a family. This is a bounded claim about a logged search, not a claim of exhaustiveness. The geometric ingredients and destination identities are classical and cited as such.

3. Definitions

Definition 3.1 (periodic Farey quotient). Start with the standard Farey graph on reduced rational numbers, with an edge between p/q and r/t when $|pt-rq|=1$, and quotient by integer translation. Denote the resulting periodic multigraph by $\mathfrak{F}$. Its vertices have unique reduced representatives $p/q \in [0,1)$, so there are exactly $\phi(q)$ vertices of denominator q, including one vertex for $q=1$. Edges are translation orbits of standard Farey edges. Parallel edge orbits are retained; loops are omitted because an edge from a vertex to itself contributes zero to the Laplacian quadratic form. All sums over incident edges in this paper count these retained edge orbits with multiplicity.

This convention is stronger than merely writing $0/1 \sim 1/1$: it fixes the edge topology needed by the Hurwitz formulas. The alternative finite interval graph with both endpoints is used only by the source $N=60$ measurement. At level $N=60$, the periodic quotient has $\Sigma_{q\leq 60}\phi(q)=1102$ vertices, while the two-endpoint interval graph has 1103. The package-local quotient constructor has 2201 distinct vertex pairs carrying 2202 translation edge orbits; the sole parallel pair is retained. **[4]** Its graph fingerprint is 0351fc9976bdbc8f9d73e8c8fd8037fa18ea84c317b0df1cdf29d9e11eb73c3e. The two operators are not silently identified.

Definition 3.2 (Ford weights). The Ford circle $B_{p/q}$ is the horoball of Euclidean diameter $1/q^2$ tangent to $\mathbb{R}$ at p/q; Ford circles are tangent iff their base points are Farey neighbors (Ford 1938). With B_∞ the standard horoball at height 1, $d(B_\infty, B_{p/q}) = 2 \log q$.

Proposition 3.3 (Ford dictionary; imported, Paper 5 Thm 1.2) [proved]. For every edge $e = \{u, v\}$ of $\mathfrak{F}$ and every $s \in \mathbb{C}$,

$$w_e(s) := e^{-s[d(B_\infty,B_u) + d(B_\infty,B_v)]} = (q_u q_v)^{-2s}.$$

Proof pointer: immediate from Definition 3.2; full statement and provenance in Paper 5, Thm 1.2. The only distance identity consumed here is $d(B_\infty,B_{p/q})=2 \log q$, obtained directly from the height ratio of the standard horoball and the Ford horoball of Euclidean diameter q^{-2}. Ford's primary article supplies the circle construction. The use of these depths as graph-Laplacian edge weights is, within the logged review, a bounded novelty claim (§2). ■

Definition 3.4 (the family and its core). Write $\sigma = \mathrm{Re}\ s$. On the core $c_c(\mathfrak{F})$ of finitely supported functions in $\ell^2(\mathfrak{F})$ define

$$(L(s)\psi)(v) = W_s(v)\psi(v) - \Sigma_{e=\{u,v\}}\, w_e(s)\, \psi(u)\,,\quad W_s(v) := \Sigma_{e\ni v}\, w_e(s)\,,$$

the row-sum-zero weighted Laplacian, with associated quadratic form $q_s[\psi] = \Sigma_{e=\{u,v\}} w_e(s) |\psi(u) - \psi(v)|^2$ (for $s \in \mathbb{R}$; for $s \notin \mathbb{R}$, $L(s)$ is defined entrywise and is non-self-adjoint, and all boundedness statements are proved through the absolute weights $|w_e(s)| = (q_uq_v)^{-2\sigma}$). On the proved bounded domain $\mathrm{Re}\ s>1/2$, $L(s)$ is self-adjoint for real s and non-self-adjoint for nonreal s. No self-adjoint realization at or left of the boundary is asserted here. Parallel edges contribute separately to each displayed edge sum.

Definition 3.5 (two truncations). Let $F_N = \{p/q \in \mathfrak{F} : q \leq N\}$ and P_N the orthogonal projection onto $\ell^2(F_N)$.

- $L_N(s)$: the Laplacian of the induced subgraph on F_N (**graph truncation**); each entry is a finite sum of terms $(q_uq_v)^{-2s}$, hence entire in s.

- $\hat{L}_N(s) := P_N L(s) P_N$ retaining the infinite-graph diagonal $W_s(v)$ (**full-weight truncation**); its diagonal carries the Hurwitz pole of Theorem 5.1.
- $L_N^{2e}(s)$: notation reserved for the source computation's two-endpoint interval graph. It is not the same finite graph as $L_N(s)$ and is used only when reporting that source measurement.

Sections 5-6 concern $L_N(s)$; the gap bracket cited in Q1 (Paper 5) requires an endpoint-convention audit before import, and the numerics of Supplementary Measurement 6.5 separately concerns $L_N(s)$ and $L_N^{2e}(s)$. None of the three matrices is silently substituted for another.

Definition 3.6 (relation to the magnetic operator). The family $L(s)$ is phase-free. Papers 1/3 define a separate magnetic Farey Laplacian and position unitary U at $s=1$. This paper neither imports an RH criterion from that construction nor defines a magnetic family away from $s=1$; any such deformation remains a separate open problem.

Notation.

Symbol	Meaning
$s, \sigma = \mathrm{Re}\ s$	family parameter; its real part
w	operator-moment variable (§6)
$\phi(q)$	Euler totient
$\Phi_{Eis}(w)$	modular Eisenstein coefficient $\Lambda(2w-1)/\Lambda(2w)$
$\Lambda(z)$	$\pi^{-z/2}\Gamma(z/2)\zeta(z)$
$\zeta(z, x)$	Hurwitz zeta $\Sigma_{t\geq 0}(t+x)^{-z}$
$\psi(x)$	digamma function (context separates it from vectors ψ)
$\mathfrak{F}$, F_N, P_N	Farey graph; level truncation; its projection
$L_N(s)$, $\hat{L}_N(s)$	periodic graph truncation; periodic full-weight truncation
$L_N^{2e}(s)$	source two-endpoint interval graph truncation

4. Existence dichotomy and the zeta norm

Lemma 4.1 (Bézout neighbor structure) [proved]. Let $p/q \in \mathfrak{F}$ with $q \geq 2$, and let $a \in \{1, \ldots, q-1\}$ satisfy $ap \equiv 1 \pmod q$. The Farey neighbors of p/q in the infinite graph have denominators exactly

$$\{a + tq : t \geq 0\} \cup \{(q-a) + t'q : t' \geq 0\},$$

the first progression listing the neighbors r/t with $pt - rq = +1$, the second those with $pt - rq = -1$; within each progression each denominator occurs for exactly one neighbor. For $q = 2$ the two progressions coincide as integer sets ($a = q - a = 1$) but enumerate distinct edge orbits (the $+1$ and -1 Bézout branches); at the cusp these are parallel edges and are retained by Definition 3.1. For the single $q = 1$ vertex, the non-loop incident edge denominators are $n \geq 2$ on each Bézout branch. At $n=2$ the two branches are parallel; for $n\geq 3$ they have distinct other endpoints. The denominator-one loop is omitted. If the two-endpoint interval convention is used instead, each endpoint has only its corresponding branch. All divergence statements below survive because they use only a tail of one branch.

Proof. If r/t is a neighbor then $pt - rq = \varepsilon \in \{\pm 1\}$, so $t \equiv \varepsilon p^{-1} \equiv \varepsilon a \pmod q$; conversely for each $t \equiv \varepsilon a \pmod q$, $t \geq 1$, the congruence $pt \equiv \varepsilon \pmod q$ determines r uniquely with $\gcd(r,t) = 1$, giving exactly one neighbor of denominator t on the branch ε. (Imported from Paper 1's neighbor lemma, lineage C-3.0.) ■

Lemma 4.2 (Hurwitz tail bound) [proved]. For real $z > 1$ and $x \in (0,1]$, $\zeta(z, x) \leq x^{-z} + \zeta(z)$.

Proof. $\zeta(z,x) = x^{-z} + \Sigma_{t\geq 1}(t+x)^{-z} \leq x^{-z} + \Sigma_{t\geq 1}t^{-z}$. ■

Lemma 4.3 (vertex-weight bound) [proved]. For $\sigma > 1/2$ and every vertex $v = p/q$,

$$\Sigma_{u \sim v} |w_{uv}(s)| = W_\sigma(v) \le 2\zeta(2\sigma)\ .$$ **[5]**

Proof. By Lemma 4.1 and $|w_{uv}(s)| = (q\ q_u)^{-2\sigma}$, for $q \ge 2$

$$W_\sigma(p/q) = q^{-2\sigma}[\Sigma_{t\ge 0}(a+tq)^{-2\sigma} + \Sigma_{t\ge 0}((q-a)+tq)^{-2\sigma}] = q^{-4\sigma}[\zeta(2\sigma,\ a/q) + \zeta(2\sigma,\ (q-a)/q)]\ .$$ **[6]**

By Lemma 4.2, each Hurwitz summand obeys (with $b \in \{a,\ q-a\}$, $1 \le b \le q-1$, so $qb \ge q \ge 2$)

$$q^{-4\sigma}\zeta(2\sigma,\ b/q) \le q^{-4\sigma}(b/q)^{-2\sigma} + q^{-4\sigma}\zeta(2\sigma) = (qb)^{-2\sigma} + q^{-4\sigma}\zeta(2\sigma) \le 2^{-2\sigma} + 2^{-4\sigma}\zeta(2\sigma)\ .$$

It remains to check $2^{-2\sigma} + 2^{-4\sigma}\zeta(2\sigma) \le \zeta(2\sigma)$, i.e. $x \le (1 - x^2)\zeta(2\sigma)$ with $x = 2^{-2\sigma} \in (0,\ 1)$. Restricting ζ to powers of two, $\zeta(2\sigma) \ge \Sigma_{k\ge 0}(2^k)^{-2\sigma} = 1/(1 - x)$, and $1/(1 - x) \ge x/(1 - x^2)$ (equivalent to $1 + x \ge x$), so $(1 - x^2)\zeta(2\sigma) \ge (1 - x^2)/(1 - x) = 1 + x \ge x$. Hence each summand is $\le \zeta(2\sigma)$, and summing the two branches gives $W_\sigma(p/q) \le 2\zeta(2\sigma)$ for $q \ge 2$. For the single denominator-one vertex, the omitted loop changes only the finite regular part:

$$W_\sigma([0]) = 2\ \Sigma_{n\ge 2}\ n^{-2\sigma} = 2(\zeta(2\sigma)-1) \le 2\zeta(2\sigma).$$ **[7]**

Under the two-endpoint interval convention each endpoint has one branch, and the same upper bound holds. No claim that the bound is attained is needed. ■

Theorem 4.4 (existence dichotomy and zeta norm; Theorem A; C-27 i-ii) [proved]. Let $s \in \mathbb{C}$, $\sigma = \mathrm{Re}\ s$.

(a) If $\sigma > 1/2$, then $L(s)$ extends from $c\_c(\mathfrak{F})$ to a bounded operator on $\ell^2(\mathfrak{F})$ with

$$\|L(s)\| \le 2\ \sup_v W_\sigma(v) \le 4\zeta(2\sigma)\ .$$

(b) If $\sigma \le 1/2$, the positive absolute-weight form q_σ is $+\infty$ for every nonzero $\psi \in c\_c(\mathfrak{F})$. In particular, for real $s \le 1/2$ the displayed positive form does not define a self-adjoint operator with form core $c\_c(\mathfrak{F})$. For nonreal s this is an absolute-convergence obstruction, not a self-adjointness statement. Realizations on other domains or in distributional frameworks are not excluded.

Proof. (a) For finitely supported $\psi,\ \chi$,

$$|\langle\chi,\ L(s)\psi\rangle| \le \Sigma_v\ W_\sigma(v)|\chi(v)||\psi(v)| + \Sigma_{u\sim v}\ (q_uq_v)^{-2\sigma}\ |\chi(v)||\psi(u)|\ .$$

The diagonal part is bounded by $\sup_v\ W_\sigma(v)\cdot\|\chi\|\|\psi\|$. The off-diagonal part is a Schur test with row / column sums W_σ: by Cauchy-Schwarz on the measure $(q_uq_v)^{-2\sigma}$ over directed edges,

$$\Sigma_{u\sim v}\ (q_uq_v)^{-2\sigma}|\chi(v)||\psi(u)| \le [\Sigma_{u\sim v}(q_uq_v)^{-2\sigma}|\chi(v)|^2]^{1/2}\ [\Sigma_{u\sim v}(q_uq_v)^{-2\sigma}|\psi(u)|^2]^{1/2} \le \sup_v\ W_\sigma(v)\ \cdot\ \|\chi\|\|\psi\|\ .$$

Hence $\|L(s)\| \le 2\ \sup_v\ W_\sigma(v) \le 4\zeta(2\sigma)$ by Lemma 4.3. **[8]** All estimates used only σ, so (a) holds for non-self-adjoint $L(s)$ ($\mathrm{Im}\ s \ne 0$) as well.

Constant audit at $s = 1$: $4\zeta(2) = 2\pi^2/3$, identical to the canonical bound $\|L\| \le 2\pi^2/3$ of Paper 1 (C-4). This closes the constant-normalization flag of packet BC-016.

(b) Let $\psi \in c\_c(\mathfrak{F})$, $\psi(v_0) \ne 0$ for some $v_0 = p/q$. All but finitely many neighbors u of v_0 lie outside $\mathrm{supp}\ \psi$, and each such edge contributes $w_{uv_0}(\sigma)|\psi(v_0)|^2$ to $q_\sigma[\psi]$. By Lemma 4.1 the incident weight sum contains $q^{-2\sigma}\ \Sigma_{t\ge T}\ (a + tq)^{-2\sigma}$ for some finite T (for $q = 1$, use either non-loop branch with denominators $n\ge 2$; under the two-endpoint convention a tail of one progression always survives, per Lemma 4.1), which diverges for $\sigma < 1/2$ (comparison with Σ t^{-2σ}), and at $\sigma = 1/2$ exactly it is a shifted harmonic series: Σ_{t≥T} (q(a+tq))^{-1} = q^{-2}Σ_{t≥T}(a/q + t)^{-1} = +∞. Hence q_σ[ψ] = +∞. This is the unit-weight blow-up mechanism of Paper 4, Thm 2.2, applied with the explicit progression weights; it covers the boundary $\sigma = 1/2$ inclusively. ■

Remark 4.5 (boundary reading; honesty note). The norm estimate diverges with $\zeta(2\sigma)$ as $\sigma \downarrow 1/2$ (the pole is reached through $2\sigma \to 1$), and the positive form over the stated core fails at $\sigma = 1/2$. The mechanism is elementary - the pole enters only through Σ n^{-2σ} = $\zeta(2\sigma)$ - so the divergence is not by itself deep. Companion Paper 3 studies a separate Mertens-motivated magnetic deficit at s=1, but no equivalence or consequence from that paper is imported here. The numerical equality of the boundary coordinate with the real part of zeta's critical line is interpretive context only; it proves nothing about zeros.

Remark 4.6 (Im s). All bounds above depend only on σ; no spectral statement for Im s $\neq$ 0 is made in this paper. Complex-s spectral slices remain an open problem outside the present real-parameter program.

5. Hurwitz diagonal and entrywise continuation

Theorem 5.1 (Hurwitz diagonal; Theorem B; C-27 iii-iv) [proved]. Let p/q $\in \mathfrak{F}$, q $\geq$ 2, and a as in Lemma 4.1. For $\sigma > 1/2$,

W_s(p/q) = q^{-4s}[ζ(2s, a/q) + ζ(2s, (q-a)/q)] , **[9]**

and for the denominator-one vertex of the periodic quotient W_s = 2(ζ(2s)-1). The subtracted entire term is the omitted loop. If m_{uv} is the number of retained parallel edge orbits between distinct vertices u and v, the off-diagonal matrix entry is -m_{uv}(q_uq_v)^{-2s} and is entire in s. Consequently:

(i) every matrix entry of L_N(s) extends meromorphically to $\mathbb{C}$, the diagonal entries with exactly one simple pole at s = 1/2 and the off-diagonal entries entire;
(ii) Res_{s=1/2} W_s(p/q) = q^{-2} for every vertex;
(iii) s $\mapsto$ L(s) is a norm-holomorphic family of bounded operators on {$\sigma > 1/2$}, and s $\mapsto$ L_N(s) is meromorphic on $\mathbb{C}$ with the single pole at s = 1/2 (fixed N).

Proof. The diagonal identity is the computation in Lemma 4.3: Σ_{t≥0}(q(a+tq))^{-2s} = q^{-4s}ζ(2s, a/q) by definition of the Hurwitz zeta, plus the mirror branch (both series converge absolutely for $\sigma > 1/2$, so the rearrangement by branches is legitimate). The classical continuation of ζ(z, x) (Apostol, Ch. 12) is meromorphic in z with one simple pole at z = 1 of residue 1 and Laurent expansion ζ(z, x) = (z-1)^{-1} - ψ(x) + O(z-1) (ψ = digamma). Substituting z = 2s: near s = 1/2,

ζ(2s, x) = (2s-1)^{-1} - ψ(x) + O(2s-1) = (1/2)(s-1/2)^{-1} - ψ(x) + O(s-1/2) ,

so each Hurwitz factor contributes residue 1/2 in s, and

Res_{s=1/2} W_s(p/q) = q^{-4·(1/2)}·(1/2 + 1/2) = q^{-2} ,

proving (ii) (the prefactor q^{-4s} is evaluated at s = 1/2; its holomorphic variation contributes nothing to the residue). For q = 1, W_s = 2(ζ(2s)-1) and Res_{s=1/2}W_s = 2·Res_{s=1/2}ζ(2s)=2·(1/2)=1=q^{-2}; the finite subtraction changes no residue.

Sanity checks (machine-verified, Appendix A.3): at s=1, W_1([0])=2(ζ(2)-1)=π^2/3-2. For q=2, ζ(2,1/2)=π^2/2, so

W_1(1/2)=2^{-4}·2ζ(2,1/2)=π^2/16, matching $2Σ\_\{t≥0\}(2(1+2t))^\{-2\}=(1/2)Σ\_\{n odd\}n^\{-2\}=π^2/16$. The earlier $π^2/8$ check was a factor-of-two error.

For (iii), fix a compact subset of {σ>1/2} and choose ε>0 so that σ−ε>1/2 there. Differentiating an edge weight introduces powers of log(q_uq_v), and (log x)^k x^{-2σ}≤C_{k,ε}x^{-2(σ-ε)} for x≥1. The Schur sums from Theorem 4.4 therefore dominate the first and second operator-norm difference quotients locally uniformly. This proves operator-norm holomorphy rather than relying only on matrix-coefficient holomorphy. We deliberately avoid the terminology “type-(A) family”: Kato’s notion concerns closed unbounded operators with constant domain, and here the operators are bounded, where holomorphy notions coincide. ■

Remark 5.2 (continuation is entrywise only). For σ ≤ 1/2 the continued matrix of L_N(s) exists (finite matrix) but the infinite matrix does not define an operator on ℓ^2 by the stated construction (Theorem 4.4(b)). The boundary obstruction is absolute summability on the stated Hilbert-space core, not entrywise algebra. The entries of the graph truncation L_N(s), by contrast, are entire and carry no pole; all pole statements in this paper refer to L_N(s). (This is why the rescaled boundary limit of §6 must be taken on the full-weight truncation L_N(s): only its diagonal carries the Hurwitz pole, so only (2s-1)L_N(s) has a nonzero limit; (2s-1)L_N(s)\→\0 since L_N’s entries are finite at s = 1/2. This refines the source-packet statement of C-28, which was phrased for L_N(s).)

6. The rescaled boundary limit

Theorem 6.1 (rescaled boundary limit; Theorem C; C-28) [proved]. Fix N ≥ 1 and use the periodic quotient convention (Definition 3.1). Then, in operator norm on ℓ^2(F_N), with the limit taken along real s ↓ 1/2,

lim_{s↓1/2} (2s-1) L_N(s) = T_N := 2·diag(q_v^{-2}) . **[10]**

(Only the real descent is stated or used in this paper; behavior under complex approach to s = 1/2 is not claimed.) The spectrum of T_N is {2/q² : 1 ≤ q ≤ N}; the eigenspace of 2/q² is the denominator shell span{δ_{p/q} : gcd(p,q) = 1}, of dimension ϕ(q). The alternative two-endpoint interval graph has one additional denominator-one vertex, so its analogous finite shell traces acquire +1; it is not the operator used in this theorem.

Proof. Diagonal: by the Laurent expansion in the proof of Theorem 5.1, for q≥2, W_s(p/q) = q^{-4s}[2(2s-1)^{-1} - ψ(a/q) - ψ((q-a)/q) + O(2s-1)] near s = 1/2, and q^{-4s} → q^{-2}, so

(2s-1)W_s(p/q) → q^{-2}·2 = 2q^{-2} as s ↓ 1/2 ,

consistent with Res_{s=1/2}W_s = q^{-2} and 2s-1 = 2(s-1/2). For q=1, W_s=2(ζ(2s)-1) gives the same limit 2 directly. Off-diagonal entries of L_N(s) are finite expressions −m_{uv}(q_uq_v)^{-2s}, bounded near s = 1/2, so (2s-1)·(off-diagonal) → 0. The matrix is finite, so entrywise convergence implies norm convergence. The multiplicity statement is the count of reduced fractions of denominator q in [0,1), which is ϕ(q). ■

Corollary 6.2 (operator powers, entire finite moments, and the ordered totient limit) [proved].

(1) Put A_N=T_N/2. For fixed N and all w ∈ ℂ,

Tr(A_N^w) = Σ_{q≤N} ϕ(q) q^{-2w}, **[11]**

Tr(A_N^{-w}) = Σ_{q≤N} ϕ(q) q^{2w}. **[12]**

Both identities use $A_N^w=\exp(w\log A_N)$, where $\log A_N$ is the self-adjoint logarithm. They are finite sums of exponentials in w and hence entire. In particular, a finite-N moment has no w-poles. Because A_N is a strictly positive finite matrix, both powers are bounded and everywhere defined for every complex w; this finite statement carries no infinite-dimensional domain or trace claim.

(2) For real $s>1/2$, put

$B_N(s):=((2s-1)/2)\hat{L}_N(s)$ and $M_N(s,w):=\mathrm{Tr}(B_N(s)^w)$.

The matrix $B_N(s)$ is positive definite: the full infinite-graph edge form, evaluated on a vector supported in F_N, can vanish only for a constant on the connected infinite graph, and a finitely supported such constant is zero. Thus $B_N(s)^w=\exp(w\log B_N(s))$ is defined and $w\mapsto M_N(s,w)$ is entire for fixed N and real $s>1/2$. Moreover, for every compact $K\subset\mathbb{C}$,

$$\lim_{s\downarrow 1/2}\sup_{w\in K}|M_N(s,w)-\mathrm{Tr}(A_N^w)|=0. \quad \textbf{[13]}$$

Indeed $B_N(s)\to A_N$ in norm by Theorem 6.1; for fixed N, the spectra remain in a compact subinterval of $(0,\infty)$ near the limit, so logarithm and exponential functional calculus are uniformly continuous for $w\in K$.

(3) For finite N, define the inverse shell matrix $H_N=A_N^{-1}=\mathrm{diag}(q_v^2)$. Its conventional finite-dimensional spectral zeta is

$$\zeta_{H_N}(w):=\mathrm{Tr}(H_N^{-w})=\mathrm{Tr}(A_N^w). \quad \textbf{[14]}$$

For the infinite shell, define the unbounded multiplication operator

$$H_\infty\delta_v=q_v^2\delta_v,$$

with domain $D(H_\infty)=\{x\in\ell^2(\mathfrak{F}):\Sigma_v\, q_v^4|x_v|^2<\infty\}$. It is positive self-adjoint and has compact resolvent. Its eigenvalue q^2 has multiplicity $\phi(q)$. This is the precise infinite object whose spectral zeta appears below; it is not a bounded inverse on all of ℓ^2.

For domain bookkeeping, also set $A_\infty\delta_v=q_v^{-2}\delta_v$. Then A_∞ is bounded, compact, positive, and injective, $\mathrm{Ran}(A_\infty)=D(H_\infty)$, and $H_\infty=A_\infty^{-1}$ only on that range. By the functional calculus, for $z\in\mathbb{C}$,

$D(H_\infty^z)=\{x\in\ell^2(\mathfrak{F}):\Sigma_v\, q_v^{4\,\mathrm{Re}\,z}|x_v|^2<\infty\}$ when $\mathrm{Re}\,z>0$,

while H_∞^z is bounded on all of $\ell^2(\mathfrak{F})$ when $\mathrm{Re}\,z\le 0$. Consequently the finite and infinite formulas have the following noninterchangeable domains:

object	definition status	bounded / everywhere-defined window	trace-class window
A_N^w, A_N^{-w}	finite matrix	every $w\in\mathbb{C}$	every $w\in\mathbb{C}$
$B_N(s)^w$, real $s>1/2$	finite positive matrix	every $w\in\mathbb{C}$	every $w\in\mathbb{C}$
$A_\infty^w=H_\infty^{-w}$	infinite multiplication operator	$\mathrm{Re}\,w\ge 0$	$\mathrm{Re}\,w>1$
$A_\infty^{-w}=H_\infty^w$	infinite multiplication operator	$\mathrm{Re}\,w\le 0$	$\mathrm{Re}\,w<-1$

Outside the displayed bounded windows, the infinite powers are unbounded operators on their functional-calculus domains. A trace is asserted only in the trace-class windows, never by meromorphic continuation alone.

(4) As $N\to\infty$ the positive-power moment series converges locally uniformly on $\{\mathrm{Re}\,w>1\}$ to

$$\Sigma_{q\ge 1}\,\phi(q)\,q^{-2w}=\zeta(2w-1)/\zeta(2w), \quad \textbf{[15]}$$

the classical totient Dirichlet series (DLMF 27.4.6; Hardy & Wright, Thm 288; Apostol Ch. 11). Convergence follows from $\phi(q)\le q$ and domination on compact subsets by $\Sigma q^{1-2\mathrm{Re}\,w}$. Therefore

$$\zeta_{H_\infty}(w):=\mathrm{Tr}(H_\infty^{-w})=\zeta(2w-1)/\zeta(2w),\ \mathrm{Re}\ w>1.$$

The trace-class condition is exact: multiplicity $\phi(q)$ makes $\mathrm{Tr}(|H_\infty^{-w}|)=\Sigma\phi(q)q^{-2\,\mathrm{Re}\,w}$, which converges exactly for $\mathrm{Re}\ w>1$.

By contrast, $\Sigma\phi(q)q^{2w}$ converges absolutely for $\mathrm{Re}\ w<-1$, where it equals $\zeta(-2w-1)/\zeta(-2w)$, and diverges for real $w\geq-1$. For the latter claim, the prime terms are $(p-1)p^{2w}$: at $w=-1$ they dominate a constant multiple of $1/p$, and for larger w the prime subseries still diverges (or its terms fail to tend to zero). It is not the right-half-plane function used here.

(5) Combining (2) and (4), for every compact $K\subset\{\mathrm{Re}\ w>1\}$,

$$\lim_{N\to\infty}\sup_{w\in K}\left|\lim_{s\downarrow1/2}M_N(s,w)-\zeta(2w-1)/\zeta(2w)\right|=0.$$

Thus the proved order is $s\downarrow1/2$ at fixed N, followed by $N\to\infty$. No reverse limit, joint limit, commutation of limits, or statement about $\lim_{N\to\infty}L_N(s)$ at fixed s near $1/2$ is claimed.

Remark 6.3 (finite-prime Eisenstein comparison). In the normalization used here, the constant term of the real-analytic Eisenstein series $E(z,w)$ for $PSL(2,\mathbb{Z})\backslash H$ is $y^w+\Phi_{Eis}(w)y^{1-w}$, where

$$\Phi_{Eis}(w)=\Lambda(2w-1)/\Lambda(2w)=\sqrt{\pi}\ \Gamma(w-1/2)/\Gamma(w)\cdot\zeta(2w-1)/\zeta(2w)$$ **[16]**

and $\Lambda(z)=\pi^{-z/2}\Gamma(z/2)\zeta(z)$ (Iwaniec, *Spectral Methods of Automorphic Forms*, 2nd ed., §3.4, beginning p. 59). Corollary 6.2 produces only the last zeta ratio, which is the finite-prime Euler factor of $\Phi_{Eis}(w)$ on $\mathrm{Re}\ w>1$; the explicit gamma and pi factor is not produced by the graph operator.

The operator-family variable reaches $s=1/2$, while the Eisenstein coefficient remains a function of the independent variable w. No equality with the full coefficient, no scattering operator, no intertwiner, and no correspondence with the Eisenstein spectral decomposition is proved. The Dirichlet-series identity is classical; the contribution claimed here is only the operator route to its finite partial sums and the ordered limit. Closest precedent for finite-graph-spectra-to-zeta limits: Friedli-Karlsson (torus graphs, arXiv:1410.8010).

6.4 Evidence-decision matrix

The results above and the computation below answer different questions. The following matrix fixes that separation before any numerical values are interpreted.

Evidence item	Evidence class	What it decides	What it cannot decide
Theorem 4.4	analytic proof	boundedness for $\mathrm{Re}\ s>1/2$, divergence of the positive form for real $s\leq1/2$, and the tracked zeta norm bound	RH, a spectral continuation across the boundary, or a finite-size transition
Theorem 5.1	analytic proof	the Hurwitz diagonal, its entrywise meromorphic continuation, and its residue	existence of the infinite operator at the boundary or a scattering correspondence
Theorem 6.1 and Corollary 6.2	analytic proof plus a classical Dirichlet-series identity	the fixed-N norm limit, shell multiplicities, exact power bookkeeping, and the ordered $s\downarrow1/2$ then $N\to\infty$ moment limit on $\mathrm{Re}\ w>1$	a reverse or joint limit, full Eisenstein coefficient, or RH implication
Package formula tests	regression checks	whether the stated finite formulas, domains, endpoint constants, and sign conventions are implemented consistently	an independent proof of the theorems

Evidence item	Evidence class	What it decides	What it cannot decide
Supplementary Measurement 6.5	single-level finite measurement	the reported eigenvalue and localization diagnostics for the two named `N=60` truncations	asymptotic scaling, a mobility edge, or transfer to the full-weight compression
Fresh same-host environment replay	computational reproducibility check	byte-identical regeneration under a newly resolved pinned environment and both normal and optimized Python modes	cross-platform or independent-investigator reproduction
Independent operator-theory and modular-forms review	open external gate	whether the analytic arguments and normalization survive specialist review	author approval or provider authorization

The main conclusions of this paper therefore rest on the analytic arguments, not on the `N=60` table. The computation can expose implementation or convention errors and preserve a finite observation; it cannot promote that observation into an operator theorem.

The order of limits is likewise part of the result rather than an incidental detail:

Route	Status in this paper	Reason
fixed `N`, real `s↓1/2` for `B_N(s)`	proved	finite-dimensional norm convergence
then `N→∞` for `Tr(A_N^w)`, locally uniformly on `Re w>1`	proved	domination by the convergent totient Dirichlet series
`N→∞` first at fixed `s` near the boundary	open	no uniform full-weight estimate is supplied
joint or path-dependent `(N,s)` limit	open	no commutation or scaling theorem is supplied
induced graph truncation substituted for the full-weight compression	forbidden	the induced matrix has no Hurwitz-pole diagonal and its rescaled fixed-`N` limit is zero

Supplementary Measurement 6.5 (two endpoint conventions at one finite level) [measured - not a theorem about `N→∞`]. Version 2 preserves a deterministic package-local replay on the periodic quotient used in Definitions 3.1 and 3.5. The historical BC-017 calculation is preserved as a separate replay of the two-endpoint interval graph. Both use `N=60`, real `s`, the induced graph truncation, and `numpy.linalg.eigh`; they differ in endpoint topology and therefore in their matrices.

`s`	periodic 1102 normalized-projector IPR	interval 1103 eigenvector IPR	periodic λ_2	interval λ_2
1.00	0.5000	0.5000	6.187×10^{-7}	6.187×10^{-7}
0.85	0.4999	0.4999	5.846×10^{-6}	5.846×10^{-6}
0.70	0.4993	0.4993	5.505×10^{-5}	5.505×10^{-5}
0.60	0.3472	0.3434	2.308×10^{-4}	2.306×10^{-4}
0.55	0.2461	0.2211	4.596×10^{-4}	4.580×10^{-4}
0.52	0.1489	0.0952	6.834×10^{-4}	6.750×10^{-4}
0.51	0.1116	0.0593	7.761×10^{-4}	7.620×10^{-4}

The periodic diagnostic is the IPR of the normalized diagonal of the spectral projector onto the numerically detected λ_2 cluster. It is basis invariant if the cluster is multiple. At the stated tolerance the detected cluster has dimension one at every sampled `s`, so it agrees with the corresponding eigenvector IPR here. The interval column is the historical single-eigenvector diagnostic and is not relabeled as a projector statistic.

The periodic result is in experiment/periodic-n60-results.json; its constructor records graph SHA-256 0351fc9976bdbc8f9d73e8c8fd8037fa18ea84c317b0df1cdf29d9e11eb73c3e. The interval values are frozen in experiment/source-comparison-1103.json, which records SHA-256 hashes for packet BC-017-critical-descent-results-20260612.md and script bc017_critical_descent_probe.py. The exact replay environment is CPython 3.12.10, NumPy 2.4.6, mpmath 1.3.0, and threadpoolctl 3.6.0. The runner sets the BLAS-related thread-count variables before importing NumPy and records OpenBLAS 0.3.31.188.0, pthreads, SkylakeX, and one active thread in the path-safe output fingerprint.

The package-local replay closes the earlier *computational* endpoint mismatch: the periodic quotient has now been evaluated under its own edge-orbit convention. It does not prove an operator relation between the two finite matrices, and their different IPR values near the boundary prohibit pointwise substitution. The remaining structural bridge is from either induced graph truncation to the periodic full-weight compression `L_N(s)`, whose diagonal includes neighbors of denominator `>N`. Those tails are largest near `s=1/2`, precisely the regime probed. The denominator-shell limit is therefore only a candidate analogy for the measured spreading trend [conjectural]. "Mobility edge" remains an analogy with Anderson localization (Evers-Mirlin): there is no disorder ensemble and no finite-size transition theorem. Finite-size scaling across `N`, complex-`s` slices, and unfolding-aware statistics remain open computational work.

6.6 Why the finite measurement does not adjudicate the theory

The measurement samples seven real parameter values on one finite denominator cutoff. It therefore supplies neither a sequence in `N` nor a controlled approximation to the full-weight compression. The two endpoint conventions also change the matrix and, near the boundary, visibly change the localization diagnostic. Those facts make the table useful as a convention-sensitive regression target and unsuitable as evidence for a universal transition.

Accordingly, no inferential arrow runs from decreasing IPR or increasing λ_2 to Theorem 6.1, to an infinite-volume phase transition, to a mobility edge, or to RH. A future numerical paper could test such arrows only after specifying multiple `N` values, a full-weight tail approximation with error bounds, stable degeneracy handling, preregistered scaling observables, and independent replay. None of those future results is assumed here.

7. Open operator questions on the real parameter interval

The family has two mathematically distinct endpoint descriptions:

- `s = 1`: the phase-free operator `L(1)`, alongside a separate magnetic construction and deficit studied in companion Papers 1/3;
- `s ↓ 1/2` at fixed `N`: the rescaled full-weight limit and totient shells; only after that limit does `N→∞` produce the positive-power moment function compared with the finite-prime factor of `Φ_Eis(w)`.

Program 7.1 (uniform operator questions; revised from ledger item BC-GAP-3d) [conjectural]. Study the real-parameter family on `s ∈ (1/2,1]` with estimates uniform in `N`, while keeping graph truncations, full-weight truncations, and the separate magnetic construction distinct. The immediate targets are operator convergence, eigenvalue counting, and moment bounds. No RH criterion or implication is asserted, and no answer to Q1-Q5 is presently shown to imply or disprove RH.

Remark 7.2 (non-genericity and falsifiability). The family is not an arbitrary interpolant: its weights are forced by the hyperbolic geometry of Ford horoballs (Proposition 3.3), the family is norm-holomorphic with a single-pole entrywise structure (Theorem 5.1), and its bounded-operator boundary is `Re s = 1/2`

(Theorem 4.4) - all [proved]. None of these facts is known to constrain zeta zeros. The criteria F1-F3 state in advance what would obstruct only the operator questions formulated below.

The program decomposes into five precisely stated questions.

7.1 Q1 (uniform gap bracket)

Paper 5 proves a gap bracket of the form $[cN^{-4}/\log N, CN^{-4}]$ at $s=1$ for its stated finite graph [proved there]. Its constants and endpoint convention must be audited before treating it as a theorem about the periodic $L_N(1)$ defined here.

Question Q1. Determine the asymptotic scale of $\lambda_2(L_N(s))$ for each fixed real $s\in(1/2,1]$. In particular, does there exist an exponent $\alpha(s)$ and explicit positive bounds, with controlled logarithmic corrections, such that $\lambda_2(L_N(s))$ is comparable to $N^{-\alpha(s)}$? The known $s=1$ result gives $\alpha(1)=4$ in this notation. The same exponent must not be assumed at other s without proof. If a uniform family of bounds exists, determine its behavior as $s\downarrow 1/2$.

[Measured] ($N=60$, single size): the periodic $L_N(s)$ replay gives $\lambda_2=6.187\times 10^{-7}$ at $s=1$ and 7.761×10^{-4} at $s=0.51$; the separate BC-017 two-endpoint replay gives 6.187×10^{-7} and 7.620×10^{-4} at the the same sampled s values. This neither determines $\alpha(s)$ nor supports monotonicity in N. Multi-level work requires $\lambda_2(s,N)$ for at least three values of N, together with an audited import of Paper 5's constants (its C-24 refinement constants are flagged for audit - not imported unaudited).

7.2 Q2 (shell perturbation theory)

Put $z=2s-1$. Near $z=0$, write

$$L_N(s)=z^{-1}T_N+R_N(s) .$$

For fixed N, the singularity of R_N at $z=0$ is removable. For $q\geq 2$, its limiting diagonal entry is

$$q^{-2}[-4\log q-\psi(a/q)-\psi((q-a)/q)],$$

and the denominator-one entry is $2(\gamma-1)$. Its limiting off-diagonal matrix entries are $-m_{uv}(q_uq_v)^{-1}$. The earlier draft observed $\psi(a/q)\sim -q/a$ but omitted the prefactor q^{-2}; that observation does not establish growth of the diagonal remainder. Uniformity remains a question about the full matrices and their edge aggregation.

Question Q2. On the real right-hand segment, is $\sup_N \sup_{0\leq z\leq\rho}\|R_N((1+z)/2)\|<\infty$ for some $\rho>0$, with $R_N(1/2)$ defined by the removable value above? Let ρ_* be the supremum of such ρ. Is $\rho_*\geq 1$, which is exactly what is needed to reach $s=1$? The prior symmetric complex disk $|z|\leq\rho$ crossed into $\mathrm{Re}\ s<1/2$ and did not express the stated real-parameter question.

7.3 Q3 (deficit definition - a well-posedness problem) and Q4 (deficit flow)

Question Q3. Papers 1/3 define an $s=1$ magnetic pairing with a position unitary U and a deficit $\delta(N)$. Is there a mathematically canonical deformation $\delta(N,s)$ on real $s\in(1/2,1]$ with (a) $\delta(N,1)$ equal to that companion quantity; (b) real-analytic dependence on s; and (c) a fully specified deformation or rigidity statement for every magnetic ingredient? Holding U and an $s=1$ vector fixed is only a candidate definition, not a canonical choice. Until Q3 is answered, Q4 and criterion F2 are not evaluable. This paper does not import any claimed RH equivalence of the companion deficit.

Question Q4 (conditional on Q3). Compute ∂_s δ(N,s) and ask whether it admits a cusp-data expression with estimates uniform in N. This is an operator-flow question only; no sign estimate or arithmetic consequence is assumed.

7.4 Q5 (positive-power moment limits)

Use B_N(s) and M_N(s,w) from Corollary 6.2. For finite N, the optional notation H_N(s):=B_N(s)^{-1} rewrites the same entire function as Tr(H_N(s)^{-w}); it supplies no new convergence result. Corollary 6.2 proves the s↓1/2 limit locally uniformly in w at fixed N and proves the subsequent N→∞ limit only at that boundary.

Question Q5. For which real s∈(1/2,1] and half-planes in w does M_N(s,w) have an N→∞ limit, locally uniformly in w and with stated uniformity in s? A locally uniform limit is holomorphic on its initial domain. Only after that is proved does it make sense to ask whether the limit has meromorphic continuation beyond that domain and how it compares with the boundary function ζ(2w-1)/ζ(2w). The first numerical task is to compare finite-N moments across several N and real s; it is not to search for nonexistent fixed-N poles.

7.5 Failure criteria (pre-registered, per Paper 4's obstruction discipline)

- **F1.** The quantity ρ_* in Q2 is proved to satisfy ρ_*<1; that specific uniform remainder route cannot reach s=1.
- **F2 (conditional on Q3).** A proof that ∂_sδ(N,s) changes sign on an interval or a positive-measure set obstructs a monotonicity-based argument. Discrete numerical sign changes are only a trigger for such a theorem and do not establish positive measure or rule out every integrated estimate.
- **F3.** For some specified real s∈(1/2,1), the sequence M_N(s,w) is proved not to converge locally uniformly on any common right half-plane. This concerns the N→∞ family, never a fixed-N trace. Any stronger continuation-based failure criterion must first name its target domain and singularity class.

A proved item would obstruct the corresponding route only. publication history and theorem status require separate review.

7.6 Scope and limitations

What is proved in this paper is finite, fixed-N, or single-operator mathematics: a dichotomy and a norm bound for the infinite operator (Theorem 4.4), entrywise continuation (Theorem 5.1), and a fixed-N rescaled limit with a classical Dirichlet-series identity recovered in a stated order of limits (Theorem 6.1, Corollary 6.2). Not proved anywhere here: any statement about lim_{N→∞} L_N(s) near s = 1/2; any spectral statement for Im s $\neq$ 0; any N → ∞ scaling of the IPR; any uniform-in-N perturbation bound; any transfer of Theorem 6.1's mechanism to either induced graph truncation used in Supplementary Measurement 6.5. The numerical evidence is from a single size (N=60), real s only, without finite-size scaling, and is labeled [measured] wherever cited. The novelty claim is bounded by the logged prior-art review and is not a claim of exhaustiveness. RH is not proved, claimed, or implied by anything here. Section 7 poses operator questions only, and F1-F3 are failure criteria only for that operator program.

Appendix A: conventions and verification

A.1 Endpoint and edge convention (q = 1). The adopted object is the periodic quotient multigraph of Definition 3.1: one denominator-one vertex, parallel edge orbits retained, and loops omitted. Consequently |F_N|=Σ_{q≤N}ϕ(q), the eigenvalue 2 of T_N has multiplicity ϕ(1)=1, and Corollary 6.2 holds exactly as displayed. The omitted denominator-one loop changes the full-weight diagonal from 2ζ(2s) to 2(ζ(2s)-1) but does not change its residue or Theorem 6.1.

The alternative two-endpoint interval graph has 1+Σ_{q≤N}ϕ(q) vertices and would add one eigenvalue 1 to A_N, hence +1 to either finite trace. It is the convention of the source measurement, not of the proved trace identities. At N=60 the counts are respectively 1102 and 1103.

A.2 Gamma-factor normalization. The exact comparison is

Φ_Eis(w)=√π Γ(w-1/2)/Γ(w)·ζ(2w-1)/ζ(2w).

The operator limit supplies only the last ratio. It does not supply the gamma/pi multiplier, and it does not identify an operator with the Eisenstein scattering operator. Source: Iwaniec, 2nd ed., §3.4, beginning p. 59.

A.3 Machine verification. The package checks the corrected endpoint regular parts at s=1: W_1([0])=π²/3-2 and W_1(1/2)=π²/16. The second value is the corrected form of the prior factor-of-two sanity check. It verifies the finite positive- and inverse-power identities at integer and complex w, the trace-class half-planes in the operator-domain table, and bounded partial sums of the classical totient ratio. The Version 2 periodic experiment is generated by experiment/run_periodic_quotient.py and pinned in experiment/periodic-n60-results.json; routine package tests verify its graph fingerprint, environment, seven-row schema, numerical envelopes, and separation from the 1103 source record without repeating the four-minute dense diagonalization.

Historical evidence class: critical-descent table - packet BC-017-critical-descent-results-20260612.md (exact filename; the bare label "BC-017" is ambiguous against the unrelated Lean-bridge file of the same number), script bc017_critical_descent_probe.py, N = 60, m=1103; Gauss-Kuzmin exclusion, bc013_transfer_operator_probe.py (r = -0.06). The correction package uses tests/operator_power_checks.py to verify both power signs exactly for integer w, check a complex w numerically, check the two N=60 vertex counts and endpoint regular parts, and compare large bounded partial sums with ζ(2w-1)/ζ(2w). Its generated output is results.json. These are bounded formula checks, not proofs of the source paper's Hurwitz-residue, finite-size, asymptotic, or RH statements.

A.4 Reproducibility ladder. Reproducibility is reported in levels so that a passing local replay is not mistaken for independent validation.

1. **Formula and package regression - complete.** Normal and optimized executions pass the same finite formula, convention, citation, manifest, and authorization guards.
2. **Fresh same-host environment replay - complete.** A disposable virtual environment was resolved from experiment/requirements.txt; two experiment runs were byte-identical to each other and to the canonical result. Normal and optimized aggregate reports were also byte-identical. Exact hashes and the native-library fingerprint are preserved in reviews/DRAFT6-CLEAN-ENVIRONMENT-REPRODUCTION.md.
3. **Independent investigator or cross-platform replay - open.** No second investigator, operating system, processor family, or BLAS implementation has yet reproduced the result.
4. **Independent specialist proof review - open.** The operator-theory and modular-forms arguments have not received external sign-off.
5. **Author and release approval - open.** No typesetting, license, metadata, DOI-version, OSF, GitHub, or other repository release follows from Levels 1 and 2.

This package has completed levels 1 and 2 only. They establish deterministic same-host regeneration, not independent scientific confirmation.

Version lineage and source-faithful correction

Public Version 1 remains preserved under Zenodo record 21231887. This successor retains its operator-family question, Ford-geometric motivation, three anchor routes, numerical donor, open-problem program, and source genealogy while correcting the mathematical and evidential defects found during review.

The principal correction concerns operator powers. For `A_N=T_N/2=diag(q_v^{-2})`, the decaying totient polynomial is `Tr(A_N^w)`, equivalently the conventional spectral zeta of the separately defined inverse shell operator `H_N=A_N^{-1}`. It is not `Tr(A_N^{-w})`. The successor also separates induced graph truncation from the full-weight compression, repairs the periodic endpoint convention and the `q=2` normalization, keeps the two variables `s` and `w` distinct, and limits the Eisenstein comparison to the finite-prime zeta ratio actually obtained.

These changes are additive and lossless at the level of the legitimate research program: incorrect formulas and implications remain traceable in the correction ledger, while the intended operator question survives in a form that can be tested. No corrected statement is backdated to the public parent. The `N=60` measurement remains supplementary provenance and cannot establish an asymptotic transition, a full-weight bridge, an Eisenstein scattering operator, or RH.

Authorial Provenance

Micah Blumberg is the author and originator of the underlying SIT research program and the Branch C hypotheses developed from it. That research spans 2011–2026, and its core concepts predate the public release of ChatGPT in November 2022. AI did not originate the underlying theory, the hypotheses, the source genealogy, or the claimed research contribution. Later AI tools were used under the author's direction for bounded mathematical formulation, verification code, source and consistency checks, and editorial refinement. Neither AI dialogue nor model agreement is cited as scientific evidence.

References

- Apostol, T. M., *Introduction to Analytic Number Theory*, Springer (1976), doi:10.1007/978-1-4757-5579-4. [Hurwitz continuation and Laurent expansion, Ch. 12; totient Dirichlet series, Ch. 11.]
- Athreya, J. S. & Cheung, Y. (2014), "A Poincaré section for the horocycle flow on the space of lattices," *IMRN* 2014(10), 2643-2690, doi:10.1093/imrn/rnt003.
- Athreya, J. S., Chaubey, S., Malik, A. & Zaharescu, A., "Geometry of Farey-Ford polygons," arXiv:1410.4908.
- Bonanno, C. & Isola, S., "Spectral analysis of transfer operators associated to Farey fractions," arXiv:0708.0686.
- Evers, F. & Mirlin, A. D. (2008), "Anderson transitions," *Rev. Mod. Phys.* 80, 1355-1417, doi:10.1103/RevModPhys.80.1355. [Mobility-edge analogy, Supplementary Measurement 6.5.]
- Ford, L. R. (1938), "Fractions," *Amer. Math. Monthly* 45(9), 586-601, doi:10.2307/2302799.
- Friedli, F. & Karlsson, A., "Spectral zeta functions of graphs and the Riemann zeta function in the critical strip," arXiv:1410.8010. [Direct precedent for finite-graph-spectra-to-zeta limits; our C-28 contributes the Farey-Laplacian route, not the destination identity.]
- Hardy, G. H. & Wright, E. M., *An Introduction to the Theory of Numbers*, OUP. [Totient Dirichlet series, Thm 288.]
- Herichi, H. & Lapidus, M., "Towards quantized number theory: spectral operators and an asymmetric criterion for the Riemann hypothesis," arXiv:1501.05362.

- Iwaniec, H., *Spectral Methods of Automorphic Forms*, 2nd ed., GSM 53, AMS (2002), doi:10.1090/gsm/053. [Eisenstein series and its Fourier expansion, §3.4, beginning p. 59.]
- Hejhal, D. A., *The Selberg Trace Formula for PSL(2,R), Volume 2*, Lecture Notes in Mathematics 1001, Springer (1983), doi:10.1007/BFb0061302. [Classical spectral / scattering context only.]
- Kanemitsu, S. & Yoshimoto, M. (1996), "Farey series and the Riemann hypothesis," *Acta Arith.* 75.
- Kleban, P. & Ozluk, A. E. (1999), "A Farey Fraction Spin Chain," *Comm. Math. Phys.* 203(3), 635-647, doi:10.1007/s002200050629; arXiv:cond-mat/9808182.
- Knauf, A. (1993), "On a Ferromagnetic Spin Chain," *Comm. Math. Phys.* 153(1), 77-115, doi:10.1007/BF02099041.
- Fiala, J., Kleban, P. & Ozluk, A. (2003), "The Phase Transition in Statistical Models Defined on Farey Fractions," *J. Stat. Phys.* 110(1-2), 73-86, doi:10.1023/A:1021014627403.
- Fiala, J. & Kleban, P. (2004), "Thermodynamics of the Farey Fraction Spin Chain," *J. Stat. Phys.* 116(5-6), 1471-1490, doi:10.1023/B:JOSS.0000041727.66775.d3.
- Prellberg, T., Fiala, J. & Kleban, P. (2006), "Cluster Approximation for the Farey Fraction Spin Chain," *J. Stat. Phys.* 123(2), 455-471, doi:10.1007/s10955-006-9034-x. [Spin-chain transition context; §2.]
- Lax, P. & Phillips, R., *Scattering Theory for Automorphic Functions*, Ann. of Math. Studies 87, Princeton (1976).
- Lewis, J. & Zagier, D. (2001), "Period functions for Maass wave forms. I," *Ann. of Math.* 153(1), 191-258.
- Marklof, J. (2010), "Horospheres and Farey fractions," arXiv:1003.4613, doi:10.48550/arXiv.1003.4613.
- Mayer, D. (1991), "The thermodynamic formalism approach to Selberg's zeta function for PSL(2,Z)," *Bull. AMS* 25(1), 55-60; Mayer (1990), "On the thermodynamic formalism for the Gauss map," *Comm. Math. Phys.* 130(2), 311-333, doi:10.1007/BF02473355.
- NIST Digital Library of Mathematical Functions, §27.4, Eq. 27.4.6. [Totient Dirichlet series and domain `Re z>2`.]
- Prellberg, T. (2003), "Towards a Complete Determination of the Spectrum of a Transfer Operator Associated with Intermittency," *J. Phys. A* 36(10), 2455-2461, doi:10.1088/0305-4470/36/10/306. [Farey-map transfer-operator context.]
- Selberg, A. (1956), "Harmonic Analysis and Discontinuous Groups in Weakly Symmetric Riemannian Spaces with Applications to Dirichlet Series," *J. Indian Math. Soc.* 20, 47-87. [Classical trace-formula and Eisenstein / scattering context only.]
- Zhang, Z. & Comellas, F. (2011), "Farey graphs as models for complex networks," *Theor. Comput. Sci.*; cf. arXiv:1201.4228.

Companion papers (Branch C series; exact published version DOIs):

- Paper 1, *The Weighted Farey Laplacian and the HSIT-Riemann Program*, doi:%6010.5281/zenodo.21231873: `C-4 bound`2π²/3,\neighbor\lemma,\Commutator-Pythagoras\identity,\and\separate\magneticU‘.
- Paper 3, *The Riemann Hypothesis as a Saturation Law*, doi:%6010.5281/zenodo.21231877: `historical`s=1‘ endpoint and deficit; no claimed RH equivalence is imported here.
- Paper 4, *Obstruction Theorems for Farey-Spectral Approaches to the Riemann Hypothesis*, doi:%6010.5281/zenodo.21231879‘: blow-up and obstruction discipline.
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- Paper 6, *M_SIT: A Lattice Field Theory Specification of Super Information Theory and the Lift Conjecture*, doi:%6010.5281/zenodo.21231885‘: SIT terminology only; no proof transfer.
- Paper 8, *Reproducible Computational Evidence for the H_SIT Program*, doi:%6010.5281/zenodo.21231891‘: computational-tier vocabulary and certificates; Version 2's package-local experiment is deterministically reproducible in the documented same-host environment, while independent cross-platform replay remains open.

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Proof packets (wiki/automation/sit-proof-program/leaves/branch-c/): BC-015-ford-circle-dictionary-20260612.md (C-26), BC-016-fordfamily-zeta-norm-20260612.md (C-27), BC-017-critical-descent-results-20260612.md (descent table), BC-018-critical-limit-totient-spectrum-20260612.md (C-28), BC-013-transfer-operator-probe-20260612.md (Gauss-Kuzmin exclusion).

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